

1. Introduction

The objective of this lab is to perform exploratory data analysis on text data collected from YouTube. Specifically, the lab focuses on comparing **video captions** and **user comments** to understand how professionally authored content differs from user-generated text.

By analyzing captions and comments from the same YouTube video, this lab aims to explore differences in text length, vocabulary diversity, and commonly used words. These comparisons help highlight how information is presented by content creators versus how viewers respond and engage with that content.

2. Tools and Setup

This lab was implemented using **Python (version 3.10+)**. The following libraries and tools were used:

- **matplotlib** – to generate histograms and visualize text length distributions
- **collections.Counter** – to compute word frequencies
- **webvtt-py** – to read and extract text from YouTube caption files (.vtt)
- **yt-dlp** – to download auto-generated English captions from YouTube

The yt-dlp tool was installed using pip and used to download the subtitle file for the selected video. All analysis was performed locally using Python.

3. Data Collection

The YouTube video selected for this lab is:

- **Title:** *The Paradox of an Infinite Universe*
- **Channel:** Kurzgesagt – In a Nutshell
- **Category:** Educational / Science
- **Upload Date:** February 27, 2024

This video satisfies the lab criteria as it is educational, non-controversial, and contains a large number of viewer comments.

- **Captions (captions.vtt)** were downloaded using yt-dlp with English auto-subtitles enabled.
- **Comments (comments.txt)** were collected manually by copying visible comments from the YouTube video page and saving them into a plain text file.

Both files were stored locally and used as input for the analysis.

4. Data Processing

4.1 Loading Comments and Captions

The comments and captions were loaded into Python using separate functions. Comments were read from a text file, while captions were extracted using the webvtt library.

Code snippet (loading comments):

```
def load_raw_comments(filepath="comments.txt"):
    comments = []
    with open(filepath, "r", encoding="utf-8", errors="ignore") as f:
        for line in f:
            line = line.strip()
            if len(line) > 0 and not line.lower().endswith("ago") and line.lower() != "reply":
                comments.append(line)
    return comments
```

Code snippet (loading captions):

```
def load_vtt_captions(filepath="captions.vtt"):
    captions = []
    for caption in webvtt.read(filepath):
        text = caption.text.replace("\n", " ").strip()
        if text:
            captions.append(text)
    return captions
```

4.2 Cleaning and Preprocessing

Only basic preprocessing steps were applied, as required for exploratory analysis:

- Empty lines & UI-related words like “reply” were removed.

- Timestamp lines such as “1 year ago” were ignored. Still some timestamp variants such as ‘1 year ago (edited)’ may remain.
- Text was converted to lowercase during analysis for consistency.

No advanced preprocessing techniques such as stemming or lemmatization were applied, since the goal of this lab was to analyze the raw characteristics of the text data.

5. Experiments and Results

This section presents the results obtained from the three required experiments conducted on YouTube caption and comment data.

5.1 Histogram of Caption vs. Comment Lengths

A histogram was generated to compare the **character lengths** of captions and comments. The plot is shown in **Figure 1**.

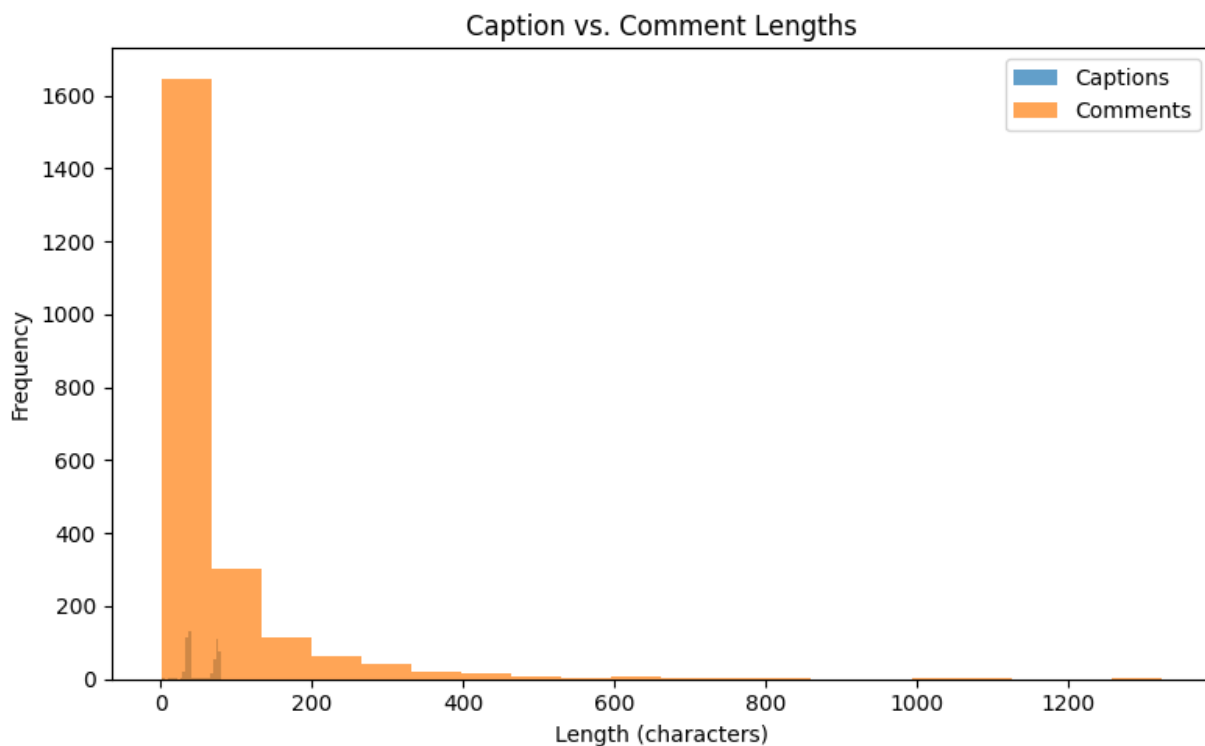


Figure 1: Histogram of caption and comment lengths (characters)

From Figure 1, it can be observed that caption lengths are relatively short and concentrated within a narrow range. This is expected, as captions are written to match spoken narration and are usually concise. In contrast, comment lengths show a much wider distribution. While many

comments are short, there are also a significant number of long comments, which creates a long tail in the histogram. This difference highlights the contrast between professionally authored text (captions) and user-generated content (comments).

5.2 Vocabulary Diversity (Type-Token Ratio)

Vocabulary diversity was measured using the **Type-Token Ratio (TTR)**, which is calculated as the number of unique words divided by the total number of words.

The computed results show that:

- **Caption TTR (0.09)** is lower than **Comment TTR (0.21)**
- **Comment TTR** is higher, indicating more diverse word usage

This result suggests that captions tend to reuse key topic-related terms, while comments contain a wider variety of words due to personal opinions, reactions, and discussion among viewers.

5.3 Top-N Word Frequency (After Stopword Removal)

After removing common English stopwords, the top 20 most frequent words were extracted separately for captions and comments.

Top caption words (sample):

- universe
- infinite
- finite
- edge
- time

Top comment words (sample):

- universe
- infinite
- space
- big
- what

The caption words are mostly technical and topic-focused, reflecting the educational nature of the video. In contrast, the comment words include more conversational and expressive terms,

showing how viewers engage with the content through questions, reactions, and personal thoughts.

No additional plots were generated, as the lab requirements were fully satisfied by the histogram and printed statistical outputs.

6. Analysis and Discussion

The experimental results clearly demonstrate differences between caption text and comment text. Captions are structured, concise, and focused on explaining concepts, which results in shorter and more consistent text lengths. Comments, on the other hand, vary significantly in length and style, reflecting the informal and spontaneous nature of user interaction.

The histogram shows that most comments are short, but a small number of very long comments significantly increase the overall range. This behavior is typical of real-world user-generated data, where some users provide detailed explanations or discussions.

Vocabulary diversity analysis further supports this observation. The higher Type-Token Ratio in comments indicates that viewers use a broader range of words, while captions reuse specific scientific terms related to the topic of the video.

Word frequency analysis also highlights this contrast. Captions emphasize explanatory terms such as “universe” and “infinite,” while comments include more personal and reaction-based words. This shows how the same content is perceived and discussed differently by the audience.

One issue encountered during the analysis was the presence of extremely long comments, which slightly skewed the histogram distribution. However, this was not removed, as it represents natural behavior in YouTube comment sections and is relevant for exploratory analysis.

8. Conclusion

In this lab, exploratory data analysis was performed on YouTube captions and comments from the same video. The results show clear differences between professionally authored caption text and user-generated comment text.

Captions were found to be shorter, more structured, and focused on explaining key concepts, while comments showed greater variation in length and vocabulary. The higher vocabulary diversity and varied word usage in comments reflect how viewers express opinions, questions, and reactions to the content.

Overall, this lab highlights the importance of understanding text characteristics before applying advanced data mining techniques. The insights gained from this analysis provide a useful foundation for future text mining and natural language processing tasks.